

In [2]: !pip install textblob

```
Requirement already satisfied: textblob in c:\users\anuj patel\appdata\local\programs\python\python310\lib\site-packages (0.19.0)
Requirement already satisfied: nltk>=3.9 in c:\users\anuj patel\appdata\local\program s\python\python310\lib\site-packages (from textblob) (3.9.2)
Requirement already satisfied: click in c:\users\anuj patel\appdata\local\programs\pyth on\python310\lib\site-packages (from nltk>=3.9->textblob) (8.3.0)
Requirement already satisfied: tqdm in c:\users\anuj patel\appdata\local\programs\pyt hon\python310\lib\site-packages (from nltk>=3.9->textblob) (4.67.1)
Requirement already satisfied: joblib in c:\users\anuj patel\appdata\local\programs\p ython\python310\lib\site-packages (from nltk>=3.9->textblob) (1.5.2)
Requirement already satisfied: regex>=2021.8.3 in c:\users\anuj patel\appdata\local\p rograms\python\python310\lib\site-packages (from nltk>=3.9->textblob) (2025.10.23)
Requirement already satisfied: colorama in c:\users\anuj patel\appdata\local\programs \python\python310\lib\site-packages (from click->nltk>=3.9->textblob) (0.4.5)
[notice] A new release of pip is available: 23.0.1 -> 25.3
[notice] To update, run: python.exe -m pip install --upgrade pip
```

In [1]: *# import libraries*
import pandas **as** pd
import matplotlib.pyplot **as** plt
import numpy **as** np
import seaborn **as** sns
from textblob **import** TextBlob *# to process natural language*

In [5]: *# Load your dataset*
data=pd.read_csv("Sentiment_Analysis .csv")
data.head(20)

Out[5]:

Sentences

- 0 The product's exceptional quality and performa...
- 1 Poor customer support and the product's subpar...
- 2 The service met standard requirements without ...
- 3 Outstanding customer service accompanied the p...
- 4 The product's failure to meet basic requiremen...
- 5 The service provided met basic needs without n...
- 6 Remarkable product quality and performance exc...
- 7 Poor customer service and a faulty product led...
- 8 The service provided was average, meeting mini...
- 9 The purchase proved delightful, with the produ...
- 10 The product's performance fell significantly s...
- 11 The service provided was acceptable, meeting b...
- 12 Impeccable product quality and performance far...
- 13 Dreadful customer support coupled with a defec...
- 14 The service rendered met basic requirements wi...
- 15 Exceptional customer service enhanced the over...
- 16 Severe product shortcomings led to considerabl...
- 17 The service was standard, meeting basic needs ...
- 18 The purchase proved delightful, offering a sup...
- 19 Disappointing product quality and poor custome...

In [11]: *# Define the sentiment analysis function*

```
def get_sentiment(text): # meaning
    analysis=TextBlob(text) # tone
    if analysis.sentiment.polarity>0:
        return "Positive"
    elif analysis.sentiment.polarity<0:
        return "Negative"
    else:
        return "netural"
```

In [12]: *# apply sentiment analysis*

```
data["sentiment"]=data["Sentences"].apply(get_sentiment)
data.head(20)
```

Out[12]:

	Sentences	sentiment
0	The product's exceptional quality and performa...	Postive
1	Poor customer support and the product's subpar...	Negative
2	The service met standard requirements without ...	Postive
3	Outstanding customer service accompanied the p...	Postive
4	The product's failure to meet basic requiremen...	Negative
5	The service provided met basic needs without n...	Postive
6	Remarkable product quality and performance exc...	Postive
7	Poor customer service and a faulty product led...	Negative
8	The service provided was average, meeting mini...	Negative
9	The purchase proved delightful, with the produ...	Postive
10	The product's performance fell significantly s...	netural
11	The service provided was acceptable, meeting b...	Postive
12	Impeccable product quality and performance far...	Postive
13	Dreadful customer support coupled with a defec...	Negative
14	The service rendered met basic requirements wi...	Postive
15	Exceptional customer service enhanced the over...	Postive
16	Severe product shortcomings led to considerabl...	Negative
17	The service was standard, meeting basic needs ...	Negative
18	The purchase proved delightful, offering a sup...	Postive
19	Disappointing product quality and poor custome...	Negative

In [13]: `data["sentiment"].value_counts()`

Out[13]:

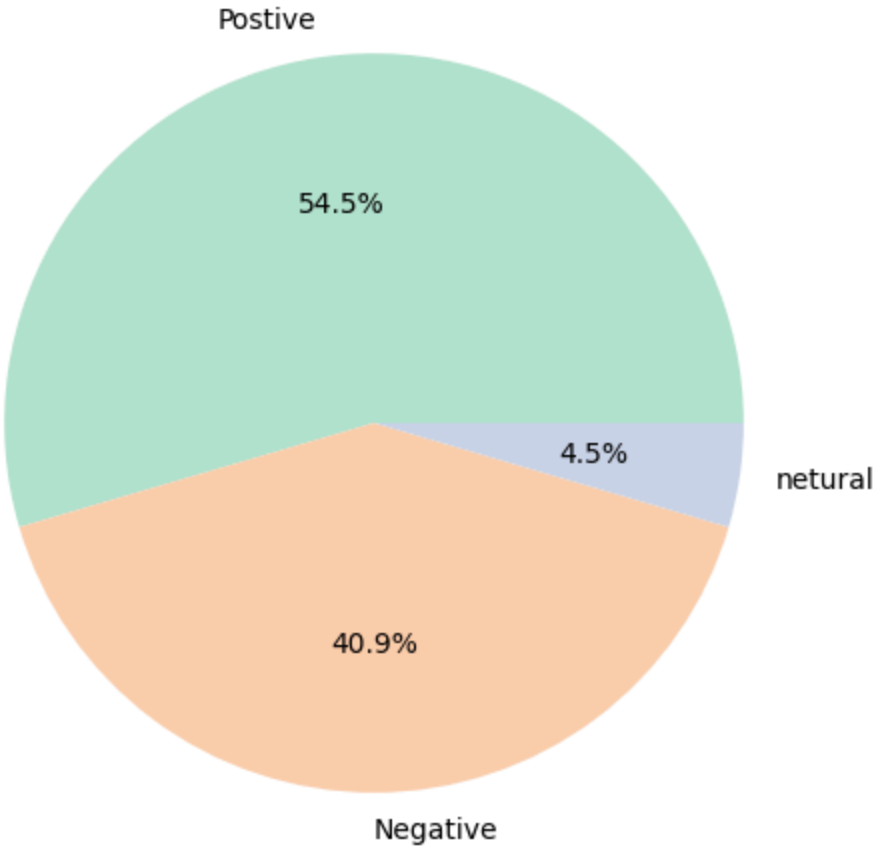
Postive	12
Negative	9
netural	1

Name: sentiment, dtype: int64

In [21]:

```
# pie chart
sentiment_counts=data["sentiment"].value_counts()
plt.figure(figsize=(8,6))
plt.pie(sentiment_counts.values,
        labels=sentiment_counts.index,
        autopct="%1.1f%%",
        colors=sns.color_palette("Pastel2"))
plt.title("Sentiment Analysis")
plt.show()
```

Sentiment Analysis



In []: